INCH-POUND

MIL-R-81498C(AS)
AMENDMENT 3
12 MARCH 1993
SUPERSEDING
AMENDMENT 2
15 May 1981

MILITARY SPECIFICATION

RECEIVING-DECODING GROUP
AN/ARA-63

This amendment forms a part of MIL-R-81498C(AS), dated 13 December 1979, and is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Paragraph 1.2 Delete in its entirety and substitute:

1.2 Classification - The equipment, covered by this specification, shall consist of the following items:

Type Designation		Applicable Paragraph
Radio Receiver	R-1379/ARA-63 R-1379A/ARA-63 R-1379B/ARA-63	3.5.1
Pulse Decoder	KY-651/ARA-63 KY-651A/ARA-63 KY-651B/ARA-63 KY-651C/ARA-63	3.5.2
Receiver Control	C-7949/ARA-63 C-7949A/ARA-63 C-7949B/ARA-63	3.5.3

AMSC N/A FSC 5826
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PAGE 1

DD Form 1426 note. Delete in its entirety and substitute:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirement Department, Code SR3, Lakehurst NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

PAGE 2

2.1 Under SPECIFICATIONS, Delete "MIL-T-5422 Testing, Environmental Aircraft Electronic Equipment." and substitute under STANDARDS, "MIL-STD-810, Environmental Test methods and Engineering Guidelines."

PAGE 3

2.1 Under STANDARDS, add the following standards:

"DOD-STD-1686, Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies, and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)."

"MIL-STD-2000, Standard Requirements for Soldered Electrical and Electronic Equipments."

2.1 Under Publications, delete "AR-5 Microelectronic Devices Used in Avionics Equipment; Procedures for selection and approval of" and substitute "MIL-STD-965, Parts Control Program."

Delete "AR-10, Maintainability of Avionics Equipment and Systems; General Requirements for" and substitute "MIL-STD-2084(AS), General Requirements for Maintainability of Avionic and Electronic Systems and Equipment."

Delete 2.1.1 and substitute: "(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)"

PAGE 4

3.2 In second and last line, delete "AR-10" and substitute "MIL-STD-2084(AS)."

3.2.1 In last line, delete "AR-5" and substitute "MIL-STD-965."

3.2.2 In second line, delete "AR-5" and substitute "MIL-STD-2084(AS)."

3.2.3 In last line, delete "AR-10" and substitute "MIL-STD-2084(AS)."

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Add as new paragraph 3.3.2.5

"3.3.2.5 Soldering - Soldering shall be in accordance with MIL-STD-2000."

Add as new paragraph 3.3.2.6

"3.3.2.6 Electrostatic Discharge - Electrostatic discharge control for the protection of electrical and electronic parts, components, assemblies and equipment shall be in accordance with DOD-STD-1686."

Paragraph 3.3.5 Interchangeability - Add the following after the last sentence:

"The equipment shall be interchangeable with a model of previously manufactured equipment supplied or designated by the procuring activity, and all assemblies, subassemblies and replaceable parts of the equipment being procured shall also be interchangeable with the corresponding assemblies, subassemblies and replaceable parts of the model of the previously manufactured equipment supplied or designated by the procuring activity."

PAGE 15

Paragraph 3.5.1.4 Contents Delete subparagraph a. (1) Local Oscillator. Add the following:

(1) Local Oscillator - The local oscillator shall be a 10-channel crystal-controlled, (one crystal for each channel), solid state unit employing multipliers, amplifiers and filters to obtain the required output.

PAGE 17

Paragraph 3.5.2.4 Contents Delete subparagraph b. (3). Add subparagraph 3.5.2.4b

(3) Provide video coding signal for modulating the IF oscillator in the Receiver BIT module. The KY/651B/ARA-63 and KY/651C Pulse Decoder BIT output circuit shall be capable of driving an impedance of 75 ohms in parallel with a 3000-pf capacitor.

PAGE 18

Paragraph 3.5.2.4 (continued)

Delete existing Subparagraph g. and substitute the following:

g. The KY-651A/ARA-63, KY-651B/ARA-63 and KY-651C/ARA-63 shall contain a glide slope selector switch for ease in glideslope adjustment.

PAGE 19

Paragraph 3.5.3.2: Delete in its entirety and substitute:

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"3.5.3.2 Form Factor - The Receiver Control shall be 5.75 inches wide by 1.50 inches high and not greater than 3.69 inches deep, not including protrusion of controls (see 3.5.3.4) and the connector."

Paragraph 3.5.3.6: Delete in its entirety and substitute:

"3.5.3.6 Panel Lighting - Panel lighting shall be as specified in MIL-P-7788. The Receiver Control panel information markings shall have uniform illumination and preclude the need for additional indicator lights for function identification; e.g., it shall not have a power ON/OFF lamp."

PAGE 20

4.2.1 Add parenthetical note "(see 6.10)."

4.2.2 In last line delete "MIL-T-5422" and substitute "MIL-STD-810."

PAGE 27

4.5.4.1 Add parenthetical note "(see 6.10)."

4.6 Add parenthetical note "(see 6.10)."

PAGE 31

6.10 Delete in its entirety and substitute:

"6.10 Consideration of data requirements. The following data requirements should be considered when this standard is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested or provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title
4.2.1	DI-NDTI-80809A	Test Reports
4.5.4.1	DI-RELI-80252	Reliability Test Reports
4.6	DI-NDTI-80807A	Test Reports

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423."

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After 6.10, add: 6.11 Subject term (keyword) listing.

Approach guidance
Cable assembly
Control signal
Distance measuring equipment (DME)
Pulse-decoder
Pulse modulation
Receiver
Signal modulation
Transmitter

6.11 Delete "NAVAIR-5334" and substitute "AIR-546X1F."

Renumber existing "6.11" as "6.12."

Preparing Activity:
Navy - AS

(Project No. 5826-N219)

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